Midterm Exam
(Solutions and Grading Criteria)
(6:35 PM – 7:50 PM : 75 Minutes)

- This exam will account for 20% of your overall grade.
- There are five (5) questions, worth 75 points in total. Please answer all of them.
- This is a *closed book*, *closed notes* exam. *No cheat sheets* are allowed.
- You are *not allowed to use your own calculator*. A scientific calculator will be available inside the Respondus Lockdown Browser.
- You are allowed to *use scratch papers* for your calculations.

Good Luck!

Question	Parts	Points
1. Asymptotic Bounds	$(a), (b), (c)$	$5+5+5=15$
2. Recurrences	$(a), (b), (c)$	$5+5+5=15$
3. Longest Increasing Subsequence	–	12
4. Selection Sort	–	18
5. Integer Multiplication	$(a), (b)$	$10+5=15$
Total		75

QUESTION 1. [**15 Points**] **Asymptotic Bounds.** For each of the following statements state whether it is *True* or *False*. You must justify each answer. Assume that all log's are of base 2.

(*a*) [**5 Points**] $\sqrt{\log n} = \mathcal{O}\left(n^{\frac{\log\log n}{\log n}}\right)$

(*b*) [**5 Points**] $4^{\sin^2 n} = \Omega\left(\log^2 n\right)$

(*c*) [**5 Points**] $2^{2n^2} = \Theta\left(2^{3n^2}\right)$

SOLUTION.

(*a*) - True: 2'. Proof: 3'

$$\begin{aligned}
\lim_{x\to\infty}\left(\frac{\sqrt{\log n}}{n^{\frac{log_c \log_c n}{\log_c n}}}\right) &= \lim_{x\to\infty}\left(\frac{\sqrt{\log n}}{c^{\log_c n \frac{\log_c \log_c n}{\log_c n}}}\right) && \because n = c^{\log_c n}\\
&= \lim_{x\to\infty}\left(\frac{\sqrt{\log n}}{c^{\log_c \log_c n}}\right)\\
&= \lim_{x\to\infty}\left(\frac{\sqrt{\log n}}{\log_c n}\right)\\
&= \lim_{x\to\infty}\left(\frac{\sqrt{\log n}}{\log n}\right)\\
&= 0 \le c && \{\text{where, } c \text{ is any positive constant}\}
\end{aligned}$$

(Alternative way: $\frac{\log_c b}{\log_c a} = \log_a b \implies n^{\frac{log_c \log_c n}{\log_c n}} = n^{\log_n \log_c n} = \log_c n$)

Hence, the given statement is *True*.

(*b*) - False: 2'. Proof: 3'

$$\begin{aligned}
\lim_{x\to\infty}\left(\frac{4^{\sin^2 n}}{\log^2 n}\right) &= \lim_{x\to\infty}\left(\frac{1}{\log^2 n}\right) \quad \because 0 \le \sin^2 n \le 1 \implies 4^0 \le 4^{\sin^2 n} \le 4^1 \implies \textit{constant}\\
&= 0
\end{aligned}$$

So, $4^{\sin^2 n} = o(\log^2 n)$

Hence, the given statement is *False*.

(*c*) - False: 2'. Proof: 3'

$$\begin{aligned}
\lim_{x\to\infty}\left(\frac{2^{2n^2}}{2^{3n^2}}\right) &= \lim_{x\to\infty}\left((\tfrac{2^2}{2^3})^{n^2}\right)\\
&= \lim_{x\to\infty}\left((\tfrac{1}{2})^{n^2}\right)\\
&= 0
\end{aligned}$$

So, $2^{2n^2} = o(2^{3n^2})$.

Hence, the given statement is *False*.

Question 2. [15 Points] Recurrences. For each of the functions below either solve it using the Master Theorem or state why the Master Theorem does not apply.

(*a*) [**5 Points**] $T(n) = \begin{cases} \Theta(1), & \text{if } n \leq 1; \\ 2T\left(\frac{n}{2}\right) + \frac{1}{2}T(n) + n^2 \log^2 n, & \text{otherwise.} \end{cases}$

(*b*) [**5 Points**] $T(n) = \begin{cases} \Theta(1), & \text{if } n \leq 1; \\ 6T\left(\frac{n}{3}\right) + (3 + \sin n)\, n^3, & \text{otherwise.} \end{cases}$

(*c*) [**5 Points**] $T(n) = \begin{cases} \Theta(1), & \text{if } n \leq 1; \\ 2T\left(\frac{n}{2}\right) + 2T\left(\frac{n}{3}\right) + \Theta(n), & \text{otherwise.} \end{cases}$

SOLUTION.

(*a*) -

$$T(n) = \begin{cases} \Theta(1) & \text{if } n \leq 1 \text{ (1)} \\ 2T(\frac{n}{2}) + \frac{1}{2}T(n) + n^2log^2n & \text{Otherwise (2)} \end{cases} \tag{0.1}$$

(2) can be simplifies as
$$\begin{aligned} T(n) - \tfrac{1}{2}T(n) &= 2T(\tfrac{n}{2}) + n^2log^2n \\ \tfrac{1}{2}T(n) &= 2T(\tfrac{n}{2}) + n^2log^2n \\ T(n) &= 4T(\tfrac{n}{2}) + 2n^2log^2n \\ &= 4T(\tfrac{n}{2}) + \Theta(n^2log^2n) \end{aligned}$$

Hence the recurrence relation can be rewrite as

$$T(n) = \begin{cases} \Theta(1) & \text{if } n \leq 1 \\ 4T(\frac{n}{2}) + \Theta(n^2log^2n) & \text{Otherwise} \end{cases} \tag{0.2}$$

Recall the Master Theorem:

$$T(n) = \begin{cases} \Theta(1) & \text{if } n \leq 1 \\ aT(\frac{n}{b}) + f(n) & \text{Otherwise } (a \geq 1,\, b > 1) \end{cases} \tag{0.3}$$

case 1: $f(n) = O(n^{log_b a - \epsilon})$ for some constant $\epsilon > 0$
$T(n) = O(n^{log_b a})$
case 2: $f(n) = \Theta(n^{log_b a} log^k n)$ for some constant $k \geq 0$
$T(n) = \Theta(n^{log_b a} log^{k+1} n)$
case 3: $f(n) = \Omega(n^{log_b a + \epsilon})$ and $af(b/n) \leq cf(n)$ for some constants $\epsilon > 0$ and $c < 1$
$T(n) = \Theta(f(n))$
From the formulations of cases in the Master Theorem, we know that case 2 applies with setting $b = 2$, $a = 4$, and $k = 2 \geq 0$ (each of a, b, k values is 1 point)

$$
\begin{aligned}
T(n) &= aT(\tfrac{n}{b}) + \Theta(n^{log_b a} log^k n) \\
&= 4T(\tfrac{n}{2}) + \Theta(n^{log_2 4} log^2 n) \\
&= 4T(\tfrac{n}{2}) + \Theta(n^2 log^2 n)
\end{aligned}
$$

Therefore, $T(n) = \Theta(n^{log_b a} log^{k+1} n) = \Theta(n^{log_2 4} log^3 n) = \Theta(n^2 log^3 n)$

2 points for the correct solution.

(b) -

$$
T(n) = \begin{cases} \Theta(1) & \text{if } n \le 1 \ (3) \\ 6T(\frac{n}{3}) + (3+\sin(n))n^3 & \text{Otherwise } (4) \end{cases} \tag{0.4}
$$

Master Theorem applies, It's the case 3. Based on the equation (4), we know $a = 6$, $b = 3$, and $f(n) = (3 + \sin(n))n^3$

Let's check if the condition of case 3 is True.

First check:

$$
\begin{aligned}
f(n) &= \Omega(n^{log_b a+\epsilon}) \qquad \text{for some } \epsilon > 0 \\
f(n) &= (3 + \sin(n))n^3 \\
f(n) &= \Omega(n^{log_3 2+\epsilon})
\end{aligned}
$$

Since $-1 \le \sin(n) \le 1$, and $n \ge 0 \implies (3-1)n^3 \le f(n) \le (3+1)n^3$ (5) (2 points)

$f(n) = \Theta(n^3)$

$$
\begin{aligned}
f(n) &= \Theta(n^{log_3 6 + log_3 \frac{3^3}{6}}) \qquad \text{Since } log_x y + log_x z = log_x yz \\
&= \Theta(n^{log_3 6 + log_3 \frac{27}{6}}) \\
&= \Omega(n^{log_3 6 + log_3 \frac{27}{6}}) \quad \text{Here, } \epsilon = log_3 \tfrac{27}{6} > 0 \text{ (1 point)}
\end{aligned}
$$

Second check: $af(\frac{n}{b}) < cf(n)$ for some $c < 1$

$$
\begin{aligned}
6f(\tfrac{n}{3}) &\le 6 \times 4(\tfrac{n}{3})^3 \quad \text{by ineqaulity (5)} \\
&= \tfrac{24}{27} n^3 \\
&= \tfrac{12}{27}(2n^3) \\
&\le \tfrac{12}{27} f(n) \qquad \text{by ineqaulity (5)}
\end{aligned}
$$

Here, we get $c = \frac{12}{27} < 1$ (1 point)

The condition holds, so $T(n) = \Theta(f(n)) = \Theta(n^3)$ (correct solution 1 point)

(c) -

$$
T(n) = \begin{cases} \Theta(1) & \text{if n} \le \text{1 (6)} \\ 2T(\frac{n}{2}) + 2T(\frac{n}{3}) + \Theta(n) & \text{Otherwise (7)} \end{cases} \tag{0.5}
$$

The Master Theorem does not apply (3 points), since equation (7) has two terms of different b's. One is 2 and the the other one is 3. (explanation 2 points)

QUESTION 3. [**12 Points**] **Longest Increasing Subsequence.** The LIS algorithm given below computes and returns the length of the *Longest Increasing Subsequence* in the input array $A[1..n]$ containing n numbers.

```
LIS ( A[1..n], n )
1.  allocate temporary array L[1..n]
2.  L[1] = 1
3.  for i = 2 to n do
4.     L[i] = 0
5.  for i = 2 to n do
6.     for j = 1 to i − 1 do
7.         if (A[j] < A[i]) and (L[j] > L[i]) then
8.             L[i] = L[j]
9.     L[i] = L[i] + 1
10. maxL = L[1]
11. for i = 2 to n do
12.    if (L[i] > maxL) then
13.        maxL = L[i]
14. return maxL
```

Analyze the worst-case running time of LIS on an input of size n.

SOLUTION.

Line	Worst Case
1.	$O(n)$
2.	$O(1)$
3.	$O(n)$
4.	$O(n)$
5.	$O(n)$
6.	$O(n^2)$
7.	$O(n^2)$
8.	$O(n^2)$
9.	$O(n)$
10.	$O(1)$
11.	$O(n)$
12.	$O(n)$
13.	$O(n)$
14.	$O(1)$

$WorstCase = O(n) + O(1) + O(n) + O(n) + O(n) + O(n^2) + O(n^2) + O(n^2) + O(n) + O(1) + O(n) + O(n) + O(n) + O(1) = O(n^2)$
Therefore, the worse case runtime is $O(n^2)$.
For lines 6-8:
Explanation: when i = 2; j = 1 to 1, i.e runs 1 times
i = 3; j = 1 to 2, i.e runs 2 times
i = n; j =1 to n-1, i.e runs n-1 times
Therefore, we have $1 + 2 + 3 + ... + n - 1 = \frac{(n-1)(n)}{2} = O(n^2)$

Grading Criteria/breakdown:
Wrong complexity for the $O(1)$s and $O(n)$s: 1 point deduction each [Max deduction is 4 points]
Complexity computation of Line 6-8: 4 points
Final summation/computation of the total worst-case running time: 4 points

Question 4. **[18 Points] Selection Sort.** Given an array $A[1..n]$ of n distinct numbers as input the *Selection Sort* algorithm rearranges the items of A in nondecreasing order of value from left (smaller array index) to right (larger array index). Recall that the outermost ***for*** loop of the algorithm iterates from $j = 1$ to $j = A.length - 1$.

Suppose the following array $A[1..10]$ is given as input to the algorithm. Please show the state of this array after each iteration of the outermost ***for*** loop.

	1	2	3	4	5	6	7	8	9	10
A	7	13	10	5	19	14	11	12	9	1

There must be one line for each value of j. Please start each line with the j value followed by a ":" and then write down the items (separated by commas, no fancy boxes are needed) of the input array in the order they will appear in $A[1..10]$ at the end of the j-th iteration of the loop.

SOLUTION.

	1	2	3	4	5	6	7	8	9	10
Input	7	13	10	5	19	14	11	12	9	1
$j=1:$	1	13	10	5	19	14	11	12	9	7
$j=2:$	1	5	10	13	19	14	11	12	9	7
$j=3:$	1	5	7	13	19	14	11	12	9	10
$j=4:$	1	5	7	9	19	14	11	12	13	10
$j=5:$	1	5	7	9	10	14	11	12	13	19
$j=6:$	1	5	7	9	10	11	14	12	13	19
$j=7:$	1	5	7	9	10	11	12	14	13	19
$j=8:$	1	5	7	9	10	11	12	13	14	19
$j=9:$	1	5	7	9	10	11	12	13	14	19

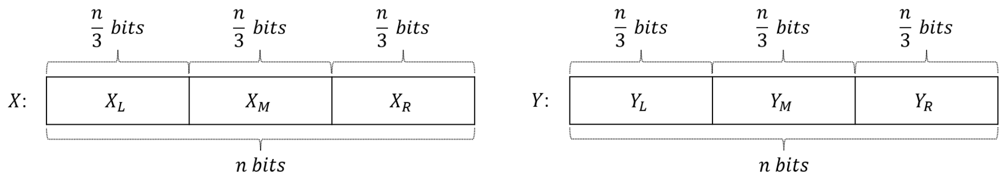
Each line (j=1, ... j=9) has 2 points.

QUESTION 5. [15 Points] Integer Multiplication. Karatsuba's algorithm multiplies two n-bit integers by recursively computing pairwise products of $\frac{n}{2}$-bit integers. This question asks you to analyze the running time of an algorithm that solves the same problem by recursively multiplying $\frac{n}{3}$-bit integer pairs.

PROBLEM: Given two n-bit integers X and Y, where $n = 3^k$ for integer $k \geq 0$, compute $Z = XY$.

ALGORITHM: If $n = 1$ then set $Z = XY$ and return. Otherwise, do the following.

Let $X = 2^{\frac{2n}{3}} X_L + 2^{\frac{n}{3}} X_M + X_R$ and $Y = 2^{\frac{2n}{3}} Y_L + 2^{\frac{n}{3}} Y_M + Y_R$.



Step 1:

$$\begin{aligned} P_0 &= X_L Y_L \\ P_1 &= X_R Y_R \\ P_2 &= (X_L + X_M + X_R)(Y_L + Y_M + Y_R) \\ P_3 &= (X_L - X_M + X_R)(Y_L - Y_M + Y_R) \\ P_4 &= (4X_L - 2X_M + X_R)(4Y_L - 2Y_M + Y_R) \end{aligned}$$

Step 2:

$$\begin{aligned} Z_4 &= P_0 \\ Z_3 &= \tfrac{P_3 - P_1}{2} + \tfrac{P_2 - P_4}{6} + 2P_0 \\ Z_2 &= \tfrac{P_2 + P_3}{2} - P_0 - P_1 \\ Z_1 &= \tfrac{P_1 + P_2}{2} - \tfrac{P_2 - P_4}{6} - 2P_0 - P_3 \\ Z_0 &= P_1 \end{aligned}$$

Step 3:

$$Z = 2^{\frac{4n}{3}} Z_4 + 2^n Z_3 + 2^{\frac{2n}{3}} Z_2 + 2^{\frac{n}{3}} Z_1 + Z_0.$$

Now answer the following questions.

(a) [**10 Points**] Write a recurrence relation describing the running time of the algorithm above when multiplying two n-bit integers. You can assume that each of $\frac{P_3 - P_1}{2}$, $\frac{P_2 + P_3}{2}$, $\frac{P_1 + P_2}{2}$, and $\frac{P_2 - P_4}{6}$ can be computed in $\Theta(n)$ time.

(b) [**5 Points**] Solve the recurrence relation from part (a).

SOLUTION.

(a) -

$$T(n) = \begin{cases} \Theta(1) & \text{if } n \leq 1 \\ 5T(\frac{n}{3}) + \Theta(n) & \text{Otherwise} \end{cases} \tag{0.6}$$

(b) -

Here $a = 5$ and $b = 3$ and $f(n) = \Theta(n)$

Therefore, $f(n) = O(n^{log_3 5-\epsilon}), \epsilon > 0$ which follows Master Theorem case 1.

So, $T(n) = \Theta(n^{log_3 5})$

Grading Criteria/Breakdown:

(a) -

Base case condition: 3 points [for $\Theta(1)$, 1.5 points and the condition $n \leq 1$, 1.5 points]

"Otherwise" condition: 7 points [$a = 5$ and $b = 3$, two points each. $f(n) = \Theta(n)$, 3 points]

(b) -

For this question, we tried to be lenient, if someone made mistakes in part A, we didn't punish for that mistake in part B. Part B was graded based on whatever value of a, b, and f(n) was obtained from part A. However, based on different values of a, b, and f(n), the Master Theorem case number may also be changed. Please note, if someone encountered Master Theorem case 3, then they also need to check the extra condition if $af(\frac{n}{b}) <= cf(n),\ c < 1$.